

MATHEMATICS**Maximum Marks: 80****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional **fifteen minutes** for **only** reading the paper.
2. You must **NOT** start writing during reading time.
3. The question paper has **9** printed pages.
4. It consists of **twenty** questions and **four** sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in **three** questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only **ONE** option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

Instruction to Supervising Examiner*Kindly read aloud the instructions given above to all the candidates present in the examination hall.***SECTION A – 20 MARKS****Question 1**

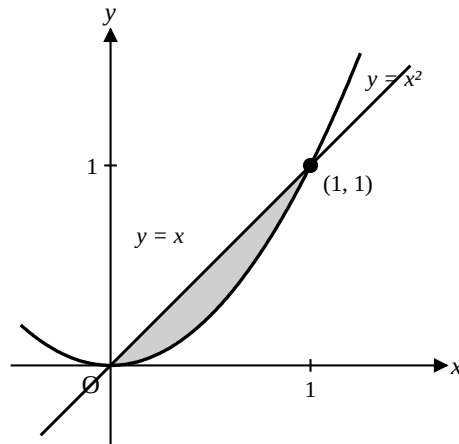
In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

- (i) The relation R on the set $\mathbb{N}$ of natural numbers defined by $a R b$ if and only if a divides b is: [1]
(Understand)
- (a) an equivalence relation
 - (b) reflexive and transitive but not symmetric
 - (c) symmetric and transitive but not reflexive
 - (d) reflexive and symmetric but not transitive

- (ii) The value of $\cot^{-1}(-1) + \operatorname{cosec}^{-1}(-\sqrt{2})$ is: (*Understand*) [1]
- (a) 0
 (b) $\frac{\pi}{2}$
 (c) π
 (d) $-\frac{\pi}{2}$
- (iii) A is a square matrix of order 3 such that $A(\operatorname{adj} A) = 5I$. Then $|\operatorname{adj} A|$ is: (*Apply*) [1]
- (a) 5
 (b) 25
 (c) 125
 (d) $\frac{1}{5}$
- (iv) The edge of a cube is increasing at 3 cm/s. How fast is its surface area increasing when the edge is 10 cm? (*Apply*) [1]
- (a) $180 \text{ cm}^2/\text{s}$
 (b) $360 \text{ cm}^2/\text{s}$
 (c) $900 \text{ cm}^2/\text{s}$
 (d) $3600 \text{ cm}^2/\text{s}$
- (v) **Statement I:** $f(x) = x + \frac{1}{x}$ is increasing on $(0, 1)$. [1]
Statement II: $f(x) = x + \frac{1}{x}$ is increasing on $(1, \infty)$.
 (*Understand*)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (vi) The slope of the normal to the curve $y = 2x^2 + 3 \sin x$ at $x = 0$ is: (*Apply*) [1]
- (a) 3
 (b) -3
 (c) $\frac{1}{3}$
 (d) $-\frac{1}{3}$
- (vii) For $f(x) = x^3 - 3x + 2$, the local maximum value and the point where it occurs are: (*Apply*) [1]
- (a) 4 at $x = -1$
 (b) 0 at $x = 1$
 (c) 2 at $x = 0$
 (d) 4 at $x = 1$

(viii) The area of the shaded region between the line $y = x$ and the curve $y = x^2$ is: (*Apply*)

[1]



(a) $\int_0^1 (x^2 - x) dx$

(b) $\int_0^1 (x - x^2) dx$

(c) $\int_0^1 (x + x^2) dx$

(d) $\int_0^1 (y - \sqrt{y}) dy$

(ix) A student evaluates $\int_0^{\frac{\pi}{2}} \sin x \cos x dx$ as follows.

[1]

Step 1: Put $t = \sin x$, so $dt = \cos x dx$.

Step 2: The integral becomes $\int_0^{\frac{\pi}{2}} t dt$.

Step 3: $= \left[\frac{t^2}{2} \right]_0^{\pi/2} = \frac{\pi^2}{8}$.

In which step is there an error (if any)? (*Understand*)

(a) Step 1

(b) Step 2

(c) Step 3

(d) There is no error

(x) Two students evaluate $\int \tan x \sec^2 x dx$. Meera writes $\tan^2 x + C$ and Arjun writes $\frac{1}{2} \sec^2 x + C$.

[1]

Whose answer is correct? (*Analyse*)

(a) Only Meera's

(b) Only Arjun's

(c) Both are correct

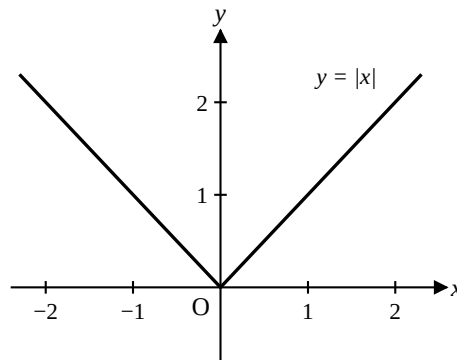
(d) Neither is correct

- (xi) The scalar projection of the vector $\hat{i} + 3\hat{j} + 7\hat{k}$ on the vector $7\hat{i} - \hat{j} + 8\hat{k}$ is: (*Apply*) [1]
- (a) $\frac{60}{\sqrt{114}}$
 (b) $\frac{60}{\sqrt{59}}$
 (c) 60
 (d) $\frac{\sqrt{114}}{60}$
- (xii) The lines $\frac{x-2}{1} = \frac{y-3}{k} = \frac{z-4}{2}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are perpendicular. The value of k is: (*Apply*) [1]
- (a) $-\frac{2}{3}$
 (b) $\frac{2}{3}$
 (c) $-\frac{3}{2}$
 (d) $\frac{3}{2}$
- (xiii) The maximum value of $Z = 3x + 2y$ subject to $x + y \leq 4$, $x \leq 3$, $x \geq 0$, $y \geq 0$ is: (*Apply*) [1]
- (a) 8
 (b) 9
 (c) 11
 (d) 12
- (xiv) **Statement I:** The feasible region defined by $x \geq 0$, $y \geq 0$ and $x + y \geq 5$ is bounded. [1]
Statement II: $Z = 2x + 3y$ has a maximum value on this region.
 (*Understand*)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (xv) If $P(A) = 0.8$, $P(B) = 0.5$ and $P(B | A) = 0.4$, then $P(A | B)$ is: (*Apply*) [1]
- (a) 0.32
 (b) 0.64
 (c) 0.4
 (d) 0.8

(xvi) **Assertion:** The function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$, whose graph is shown, is not onto. [1]

Reason: f is not one-one, since $f(-1) = f(1)$.

(Analyse)



- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xvii) **Assertion:** $\int_{-1}^1 x^2 dx = 2 \int_0^1 x^2 dx$. [1]

Reason: $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ for every continuous function f .

(Understand)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xviii) Give an example of two non-zero 2×2 matrices A and B such that $AB = O$. (Understand) [1]

(xix) Form the differential equation of the family of curves $y = A e^{2x} + B e^{-2x}$, where A and B are arbitrary constants. (Apply) [1]

(xx) X takes the values 0, 1, 2, 3 with $P(X = 0) = 0.1$, $P(X = 1) = k$, $P(X = 2) = 2k$ and $P(X = 3) = 0.3$. Find $E(X)$. (Apply) [1]

SECTION B – 14 MARKS

Question 2

Find the vector and Cartesian equations of the line through the point $(1, 2, -4)$ and perpendicular to each of the lines $\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$. [2]

(Apply)

Question 3

A speaks the truth in 75% of cases and B in 80% of cases. They are asked independently about the same fact.

- (i) Find the probability that they contradict each other. *(Apply)* [1]
 (ii) Find the probability that they agree with each other. *(Evaluate)* [1]

Question 4

Solve the differential equation $(1 + e^{2x}) dy + (1 + y^2) e^x dx = 0$, given that $y = 1$ when $x = 0$. *(Apply)* [2]

Question 5

If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, $-1 < x < 1$, $x \neq y$, prove that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$. *(Apply)* [2]

OR

The table gives values of f , g and their derivatives. If $h(x) = f(g(x))$, find $h'(1)$ and $h'(2)$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	2	2	5
2	1	-1	3	2
3	2	4	1	-3

- (a) Find $h'(1)$. *(Apply)* [1]
 (b) Find $h'(2)$. *(Apply)* [1]

Question 6

Prove that $\tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1}x$, $-\frac{1}{\sqrt{2}} \leq x \leq 1$. *(Analyse)* [2]

OR

Solve for x : $\sin^{-1}(1-x) - 2 \sin^{-1}x = \frac{\pi}{2}$. *(Apply)* [2]

Question 7

If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$, find the angle between $\vec{a}$ and $\vec{b}$. *(Analyse)* [2]

OR

Show that the points $A(1, -2, -8)$, $B(5, 0, -2)$ and $C(11, 3, 7)$ are collinear, and find the ratio in which B divides AC . *(Apply)* [2]

Question 8

A constant force $\vec{F} = 4\hat{i} + \hat{j} + 3\hat{k}$ newton moves a small robot from the point $A(1, 2, 3)$ to the point $B(5, 4, 1)$, distances in metres.

- (i) Find the work done by the force. *(Apply)* [1]
 (ii) Find the component of the force along the displacement. *(Apply)* [1]

SECTION C – 21 MARKS

Question 9

A dietician mixes two foods F_1 and F_2 . One unit of F_1 contains 2 units of vitamin A and 1 unit of vitamin B; one unit of F_2 contains 1 unit of vitamin A and 2 units of vitamin B. The mixture must contain at least 8 units of vitamin A and 10 units of vitamin B. F_1 costs ₹5 and F_2 costs ₹7 per unit.

- (i) Formulate the problem as an LPP to minimise the cost. *(Apply)* [1]
- (ii) Draw the feasible region and find its corner points. *(Apply)* [1]
- (iii) Find the minimum cost and justify that it is the minimum. *(Evaluate)* [1]

Question 10

If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$, find $\frac{d^2y}{dx^2}$. *(Apply)* [3]

Question 11

Let $y = (\log x)^x + x^{\log x}$, $x > 1$.

- (i) Find $\frac{dy}{dx}$. *(Apply)* [2]
- (ii) Find the value of $\frac{dy}{dx}$ at $x = e$. *(Apply)* [1]

Question 12

Find the coordinates of the image of the point (1, 3, 4) in the plane $2x - y + z + 3 = 0$. *(Apply)* [3]

OR

Two drones fly along the straight paths $L_1: \frac{x}{1} = \frac{y-1}{1} = \frac{z-2}{1}$ and $L_2: \frac{x-3}{2} = \frac{y-1}{-1} = \frac{z-6}{3}$.

- (a) Show that the paths intersect. *(Analyse)* [1]
- (b) Find the point where the drones could collide. *(Apply)* [1]
- (c) Find the angle between the paths. *(Apply)* [1]

Question 13

Solve for x , given $a \neq 0$: $\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$. *(Analyse)* [3]

OR

If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 4A + 3I = O$. Hence find A^{-1} . *(Apply)* [3]

Question 14

f is a continuous even function on $[-3, 3]$ with $\int_0^3 f(x) dx = 6$ and $\int_1^3 f(x) dx = 4$.

- (i) Find $\int_{-3}^0 f(x) dx$. *(Understand)* [1]

- (ii) Find $\int_{-1}^1 [f(x) + x^3 + 2] dx$. (Apply) [2]

Question 15

Consider the region $R = \{(x, y) : \frac{x^2}{16} + \frac{y^2}{9} \leq 1 \leq \frac{x}{4} + \frac{y}{3}\}$.

- (a) Sketch R and find the points where the line meets the ellipse. (Understand) [1]
 (b) Using integration, find the area of R . (Apply) [2]

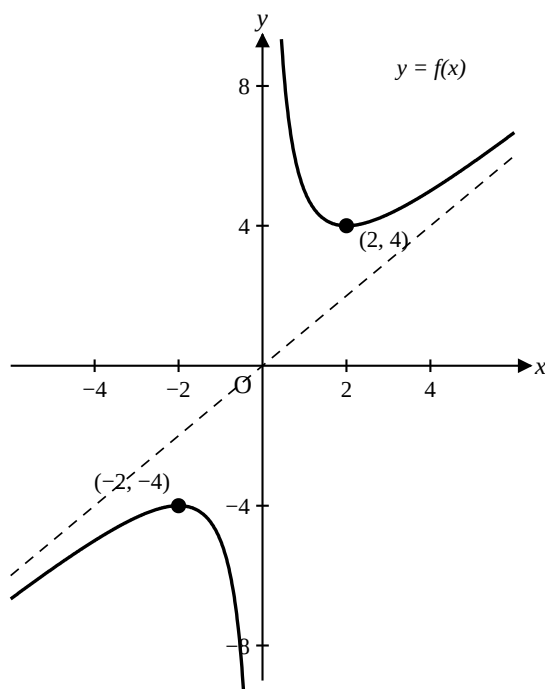
OR

Using integration, find the area of the region bounded by the curve $y = \log x$, the x -axis and the line $x = e$. (Apply) [3]

SECTION D – 25 MARKS

Question 16

The daily running cost (in thousands of rupees) of a delivery van travelling at x (in tens of km/h) is modelled by $f(x) = x + \frac{4}{x}$. The graph of f is shown.



- (i) Write the natural domain of f . (Recall) [1]
 (ii) Show that f is not one-one on $(0, \infty)$. (Understand) [1]
 (iii) Find the range of f on $(0, \infty)$, and the speed at which the cost is least. (Analyse) [1]
 (iv) $g : [2, \infty) \rightarrow [4, \infty)$ is defined by $g(x) = f(x)$. Find $g^{-1}(x)$. (Apply) [1]
 (v) Verify that $g(g^{-1}(5)) = 5$. (Apply) [1]

Question 17

Sand pours from a pipe at the rate of $12 \text{ cm}^3/\text{s}$ and forms a conical pile whose height is always one-sixth of the radius of its base.

- (a) Express the volume V of the pile in terms of its height h . (Understand) [1]

- (b) How fast is the height increasing when the height is 4 cm? (*Apply*) [2]
 (c) How fast is the radius increasing then? (*Apply*) [1]
 (d) How fast is the area of the base increasing then? (*Analyse*) [1]

OR

A closed cylindrical can is to hold a fixed volume V of liquid. Its base radius is r and its height is h .

- (a) Express the total surface area S as a function of r . (*Understand*) [1]
 (b) Show that S is least when $h = 2r$. (*Analyse*) [2]
 (c) If $V = 128\pi \text{ cm}^3$, find the dimensions of the can and its least surface area. (*Apply*) [2]

Question 18

Answer the following.

- (a) Show that $\int \sec x \, dx = \log |\sec x + \tan x| + C$. (*Understand*) [2]
 (b) Hence, evaluate $\int \frac{dx}{\sqrt{3} \cos x + \sin x}$. (*Evaluate*) [3]

OR

- (a) Show that $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$. (*Understand*) [2]
 (b) Hence, evaluate $\int \sqrt{3 - 2x - x^2} \, dx$. (*Evaluate*) [3]

Question 19

Bag I contains 5 red and 3 black balls; Bag II contains 4 red and 6 black balls. One ball is drawn at random from Bag I and put into Bag II without looking. A ball is then drawn at random from Bag II.

- (a) Find the probability that the ball drawn from Bag II is red. (*Apply*) [2]
 (b) The ball drawn from Bag II is red. Find the probability that the transferred ball was black. (*Analyse*) [2]
 (c) Which is more likely, that the transferred ball was red or black? Justify. (*Evaluate*) [1]

OR

In a game a player throws a fair die. If a six turns up he wins ₹8. If not, he throws again, and wins ₹4 if a six turns up then; otherwise he loses ₹3. Let X be his gain (in ₹).

- (a) Write the possible values of X . (*Understand*) [1]
 (b) Find the probability distribution of X . (*Apply*) [2]
 (c) Find $E(X)$. Is the game favourable to the player? (*Evaluate*) [2]

Question 20

The number of visitors $N(t)$ (in hundreds) at a new heritage museum, t months after it opened, is modelled by $N(t) = at^2 + bt + c$. Records show $N(1) = 8$, $N(2) = 15$ and $N(3) = 24$. Set up a matrix equation, solve it using the inverse of the coefficient matrix, and hence find the month in which the number of visitors first reaches 4800. (*Evaluate*) [5]

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